

Chapter 5 Newton's Laws of Motion

5.1 The First Law

A body continues in its state of rest or uniform motion in a straight line unless acted upon by an external unbalanced force. This tendency is called inertia and it depends only on the mass of the body.

The net force equals mass times acceleration:

where F is measured in newtons and a in metres per second squared.

Quantity	Symbol	Unit
Force	F	N
Mass	m	kg
Acceleration	a	m/s^2

5.2

$$F = m$$